

Project Title

The Electric Step: Harnessing Foot Traffic to Generate Renewable Energy

Author's name

Blessed Kutyaauripo – Mississippi Valley State University, 14000 Highway 82 W, Itta Bena MS

Contact details

Email: blessed.kutyaauripo@mvsu.edu

ABSTRACT

The African continent is rich in a diverse number of energy sources, including hydro energy and solar energy, but it is seemingly facing energy poverty due to unexploited energy resources. This may be due to methods of exploiting these resources being too expensive. The goal of this engineering project was to design, program and test a low-cost smart floor tile, named The Electric Step, that produces sustainable energy. The efforts included 3D printing a tile using a polycarbonate filament, then setting up lead zirconate titanate piezoelectric ceramics in a series circuit beneath the tile. The tile converted the dynamic energy from any form of mechanical stress (typically human footsteps) into chemical energy, which could be stored in a 9 V lithium polymer battery. The system was based on an ESP32 microcontroller (responsible for transmitting data over Wi-Fi) and included sensors to measure voltage and current output from the smart tile. All components were to be linked remotely on a web application, to monitor foot traffic as well as carry out arithmetic operations. The hypothesis tested was that a single tile of The Electric Step can produce 12 V of electricity every day, which would be sufficient to power an LED light bulb and charge a cell phone in a single house in Zimbabwe's high-density suburbs, such as Dzivarasekwa in the city of Harare. The results demonstrated the potential of exploiting low-cost energy in the long-term through the proposed project. The theoretical voltage of 12 V and the tested voltage of 11.4 V showed the system's ability to generate near-expected voltages, thus proving its potential for practical use since 11.4 V can still power lights and other small devices in a community like Dzivarasekwa. Furthermore, in the long term, smart floor tiles promise to cut down on high energy tariffs as they are estimated to cost \$0.10 per kWh, which is \$0.05 lower than the \$0.15 per kWh on traditional energy sources currently in use. However, for future adjustments, the practical and commercial implementation of the smart energy system would require certain security measures like the use of surveillance cameras to curb theft issues.

Definition of Keywords

- **Arduino:** *An open-source electronics platform based on easy-to-use hardware and software.*
- **Coercive field:** *A measure of the ability of a ferromagnetic material to withstand an external magnetic field without becoming demagnetized.*
- **Electric dipole moment:** *A measure of the separation of positive and negative electrical charges within a system; that is, a measure of the system's overall polarity.*
- **ESP32:** *low-power system on a chip-microcontroller with integrated Wi-Fi and dual-mode Bluetooth.*
- **LCD:** *Liquid Crystal Display; a type of flat panel display that uses liquid crystals in its primary operation.*
- **Paradigm:** *A pattern or model.*
- **Piezoelectricity:** *The ability of certain materials to generate alternating current voltage when subjected to mechanical stress.*
- **Piezoelectric ceramic:** *A smart material that converts a mechanical effect (such as pressure, movement, or vibration) into an electrical signal and vice versa.*
- **Piezoelectric effect:** *The induction of an electric charge in response to an applied mechanical strain.*
- **PZT:** *Lead zirconate titanate.*
- **Serial monitor:** *A tool used when creating Arduino projects; it can be used as a debugging tool, to test out concepts, or to communicate directly with the Arduino board.*
- **Tensile stress:** *The force per unit area of a material that is stretched by an applied load.*

INTRODUCTION

Africa is regarded as a continent blessed with a variety of natural resources (coal, crude oil and many others) that may be used to produce energy [1]. However, many communities across the continent continue to experience energy poverty. Energy poverty refers to a state of unavailability and unreliability of primary energy services (e.g. electricity, heating and cooking) to communities at affordable prices. Hydroelectricity and fossil fuels are Africa's primary energy sources, but they are proving to be very expensive to exploit due to factors like high population, often associated with very low income. A high population with low income intensifies the strain on existing energy systems which demand expensive upgrades and expansions. This also makes it challenging to provide energy at an affordable price for all. According to the *"Atlas of Africa Energy Resources"*, Africa consumes oil the most (42%) out of its energy sources, followed by gas (28%), coal (22 %), hydro energy (6%), renewable energy (1%) and nuclear energy (1%) [2]. Nigeria's population is estimated to be around 200 million, with most people living in poverty as the majority of Nigeria's citizens live below \$1 a day [3].

The cost of medium-sized hydropower plants (10 megawatts to 100 megawatts) can range from \$1,000 to \$5,000 per kilowatt, and larger projects can be more expensive due to the need for infrastructure construction and environmental impacts [4]. Furthermore, when it comes to exploiting fossil fuels, offshore oil production projects can cost between \$5 billion and \$20 billion, depending on the size and complexity of the project [5]. Building pipelines can cost between \$1 million and \$2 million per kilometer. These large expenditures put the usage of public funds into jeopardy or require heavy borrowing, which deprives the government of resources for essential sectors such as healthcare, education and infrastructure. The public is affected through increased energy tariffs or lack of access to essential public services, hence widening the differences in socio-economic status and perpetuating endless cycles of poverty [6]. Poverty can be further increased by the disintegration of the community due to migration caused by local unemployment [7].

Energy plays a very important role in the development of an economy and livelihood of the general public, which is why it is crucial to develop UN Sustainable Development Goal (SDG) 7: Affordable and Clean Energy. Nearly 60% of refrigerators used in health clinics in Africa have unreliable electricity, compromising the safe storage of vaccines and medicine [8]. The availability of energy would allow students to study and businesses to remain open for longer hours: this would lead to increased national productivity, which accelerates economic growth.

The diverse types of energy sources that could be harnessed in some parts of Africa show varying potential energy capacity and average cost of energy generated (Table 1) [9], [10], [11], [12] Solar energy, which Africa has in abundance, could provide around 10,000 gigawatts (10 terawatts) of new energy [13].

Table 1 – Data on potential energy capacity that could be harnessed in some parts of Africa as per energy source.

Energy type	Potential Capacity (GW)	Main Country	Average cost of energy generated (\$)
Geothermal	15	Ethiopia, Kenya	1.5 tn
Wind	100	South Africa	5.5 bn
Coal	300	Botswana, Mozambique, South Africa	6 bn
Hydropower	350	DRC	20 bn
Gas generated power	400	Mozambique, Nigeria, Tanzania	30 bn
Solar	10,000	All	12 tn

Even though there are abundant energy resources in Africa, most of the potential energy is not being exploited (Table 1). Some urban communities across sub-Saharan Africa do not have access to electricity. Evidence shows that at least 110 million of the 600 million people still living without access to electricity in Africa live in urban areas, and the remainder, some 80 percent, reside in rural areas [14]. The major regions where electricity does not reach include Nigeria, Zimbabwe Tanzania, Ghana and Kenya [15].

To gain insight into the practical implications of energy poverty, a field investigation was conducted in Dzivarasekwa, a high-density suburb situated on the northwestern side of the City of Harare, Zimbabwe. A section of the suburb had been completely deprived of power at a time when the country was (and still is) facing an energy crisis, which was evident in electricity being supplied to homes for a limited number of hours. Structured and semi-structured interviews were carried out to obtain information on the issue. According to the affected families, there was a fault in their main power transformer (Fig. 1) which affected at least 20 houses, leaving them with no electricity from January 5, 2023 to February 25, 2023.



Figure 1 – Burnt down transformer in Dzivarasekwa. The fault in the main power transformer left at least 20 houses with no electricity for almost two months.

The electricity problem in Dzivarasekwa is one among other examples of energy poverty in some parts of Zimbabwe and this should be improved by adopting more sustainable energy methods. Due to its flexibility, design, and structural retrofitting capability, The Electric Step is a potential solution to these problems. Since 3D printing has become more widely used, it would be cheaper for future uses in roads and buildings: the smart tiles could easily be retrofitted to existing constructions or planned into any new ground surface framework using 100% reused materials. For a household, an alternative would be using compacted recycled plastic to make the smart tiles, since less friction is experienced in homes compared to streets and pavements. If each area has a designated design of the power grid, then power surges will be minimized.

Discovered in 1880 by French physicists Jacques and Pierre Curie [16], piezoelectricity has subsequently been exploited in several useful applications, such as the production of piezoelectric inkjet printing and Braille machinery. However, as much as all of these may be breakthroughs, there remains a great need to produce more sustainable energy: thus, The Electric Step is an idea the researchers of this paper have developed to combat problems of energy poverty in Africa.

With respect to Zimbabwe's power ratings, 12 V is the standard voltage that can power light bulbs and charge a cell phone in a single house [16], therefore the aim of this project was to design a single tile of The Electric Step that could produce 12 V of electricity every day. They have managed to specially design a floor tile based on a web application, interlinked with multiple sensors and fitted with sensitive piezoelectric ceramics. The ceramics create an electric charge when stepped on or when a force is applied to the tile, and the produced energy is put away in a 9 V lithium polymer battery. A piezoelectric ceramic is a smart material that converts a mechanical effect (such as pressure, movement, or vibration) into an electrical signal [17]. Piezoelectric ceramics belong to a group of crystals which are polar without an electric field being applied: this means that these crystals have a positive charge on one side and a negative charge on the other, all of which is not influenced by the presence of any electric field.

Piezoelectric crystals are electrically neutral. The atoms inside them may not be symmetrically arranged, but their electrical charges are perfectly balanced – a positive charge in one place cancels out a negative charge nearby. However, if a piezoelectric crystal is squeezed or stretched, the structure becomes deformed, pushing some of the atoms closer together or further apart, upsetting the balance of positive and negative, and causing net electrical charges to appear (Fig. 1) [18].

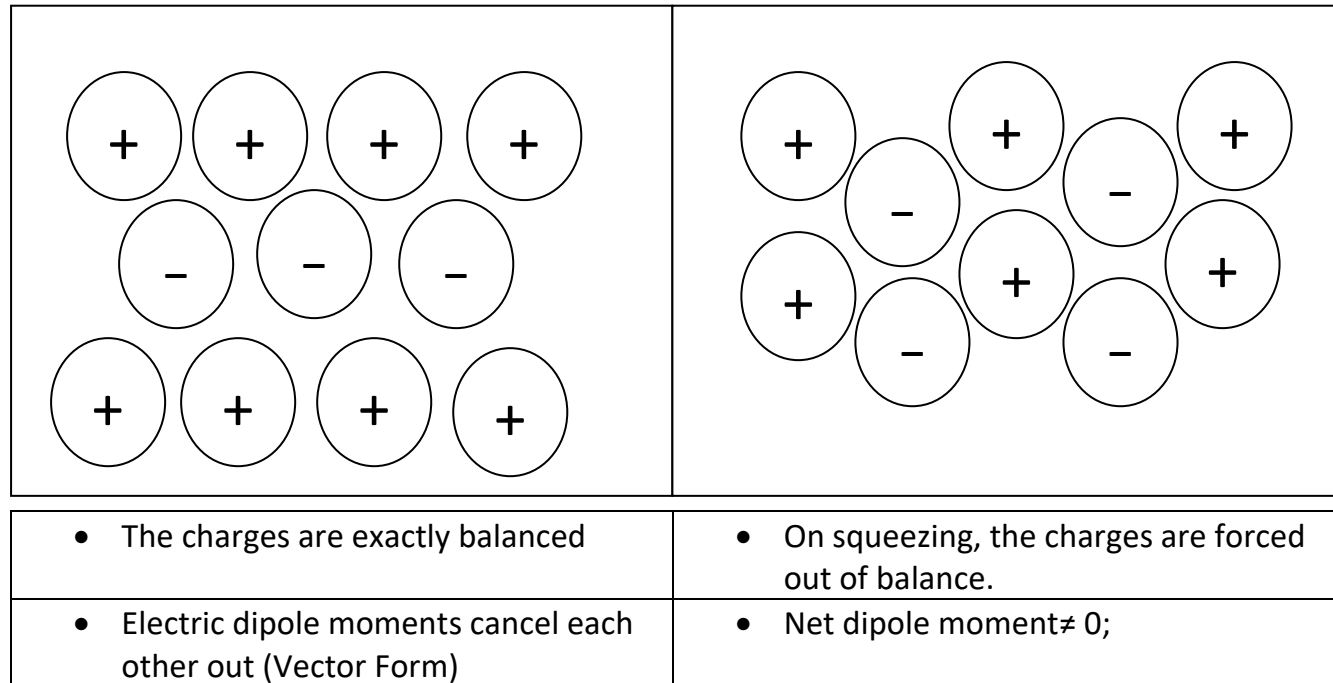


Figure 2 – Structure of the arrangement of piezoelectric crystals.

The tested results of the smart floor tile system produced an average of 11.4 V, a 0.6 V difference from the calculated theoretical voltage result. Although these results did not exactly match the calculated 12 V, The Electric Step still shows great potential to achieve the intended goal of providing the basic energy needs for a single house in Zimbabwe. Managing to provide energy for these basic needs is a major step towards decreasing the energy crisis in Zimbabwe. Over the years, piezoelectricity-related technologies to provide sustainable energy have

emerged and most of them are seemingly promising. While most of these technologies are focused on replacing already existing energy sources, The Electric Step is focused on retrofitting to already existing renewable energy technologies and collaborating with existing energy companies. Furthermore, other piezoelectricity technologies are focused solely on harnessing energy for small device applications, which is a limitation that The Electric Step will overtake by using the Internet of Things (IoT) and microcontroller systems to harness more energy for larger devices and power grids.

METHODS AND MATERIALS

Theoretical measurements and calculations

To validate the hypothesis tested, measurements of materials and tests focusing on both theoretical predictions and practical performance were crucial. The piezoelectric effect (the induction of an electric charge in response to an applied force) is common in piezo ceramics like polyvinylidene fluoride (PVDF), barium titanate (BTO) and lead zirconate titanate (PZT) [17]. Encompassing these, there are two separate types of piezoelectric effects known to date: direct effect and inverse effect. In the direct piezoelectric effect, a material is polarized and generates voltage when it is compressed, or when tensile stress is applied. This relationship can be described by the equation:

$$P=d\sigma$$

Where **P** is the polarization (electric charge per unit area), **d** is the piezoelectric coefficient (a material-specific constant), and **σ** is the applied mechanical stress [18]. This equation was used to estimate the voltage output based on the stress expected from foot traffic.

In the inverse piezoelectric effect, the application of electric potential causes mechanical displacement in the material. The equation for this effect can be expressed as:

$$\epsilon=dE$$

Where **ε** is the strain (deformation of the material), **d** is the piezoelectric coefficient, and **E** is the applied electric field [19].

According to whether the stress applied to the material is parallel or perpendicular to the electric field supplied to the polymer, the piezoelectric material can operate in either of its two principal operating modes: parallel mode (Mode 33) or perpendicular mode (Mode31) [20]. There is a coupling coefficient (**K**) for any one of the operating modes which expresses the efficiency of the electromechanical

transduction [21]. In parallel mode, the stress applied is aligned with the direction of the electric field. The efficiency of energy conversion in this mode is quantified by the coupling coefficient K_{33} , which is calculated using the equation:

$$K_{33} = \frac{d_{33}}{\sqrt{S_{33} \epsilon_{33}}}$$

Here, d_{33} represents the piezoelectric coefficient, indicating how much electric charge is generated per unit of stress; S_{33} is the material's compliance, showing how much it deforms under stress; and ϵ_{33} is the permittivity, measuring how easily the material becomes polarized by the electric field [22].

In the perpendicular mode, where stress is applied perpendicular to the electric field, the efficiency is represented by the coupling coefficient K_{31} . The coefficient is given by the equation:

$$K_{31} = \frac{|d_{33}|}{\sqrt{S_{11} \epsilon_{33}}}$$

Where $|d_{33}|$ is the absolute value of the piezoelectric coefficient, S_{11} is the material's compliance in a different direction, and ϵ_{33} is the permittivity [23]. These equations helped in selecting the appropriate piezoelectric material by evaluating its efficiency. A high coupling coefficient indicates better energy conversion efficiency.

Since piezoelectricity has many forms, the selection criteria of the materials for different situation applications are important. Many factors such as stress level applied, softness or hardness of material and disc thickness had to be considered. Data was compiled from the *IOP Conference Series: Materials Science and Engineering* (Table 2), where extensive experimentation was conducted on various piezoelectric materials [24]. While the researchers of *The Electric Step* did not conduct the experimental work of testing the piezoelectric materials, they chose this dataset due to its thoroughness and relevance to the current analysis.

Table 2 – Performance parameters of commonly used piezoelectric materials.

Parameter	Inorganic								
	Single crystal					Bi crystal		OrganicPVDF	CompositeP (VDF-TrFE)
	Quartz crystal	PZT-4	PZT-5 A	PZT-5 H	PZT-8	ZnO	AlN		
Dielectric constant, K	0.06	0.89	0.95	0.1	0.8	–	–	14	0.25
Piezoelectric strain coefficient, d_{31}	−0.67	−123	−171	−274	−97	−5	−5.43	−20	−25
Piezoelectric strain coefficient, d_{33}	2 3	289	374	593	225	12.4	5	30	20
Young’s modulus, E_T	4.6	38	50	100	29	–	8.5	12 15	10–15
Dielectric loss, $\tan\delta$	–	0.004	0.02	0.02	0.004	–	–	–	0.018
Mechanical quality factor, Q_m	100	500	75	600	1000	1770	2490	3500	0.018
Elastic modulus, E (GPa)	77.8	76.5	61	49	--	210	310	2 8	2.3
Poisson’s ratio, ν	0. 17	0.32	0.31	0.36	0.3	0.33	0.25	0.33	0.4
Density, ρ (kg m ^{−3})	2650	7500	7750	7500	7600	5660	3260	1750	890

An analysis of various commonly used piezoelectric materials from the list above was made [25], [26]. The choice of piezoelectric material was largely dependent on the coupling coefficient k and the consideration of factors like stress levels, energy needs and cost. The researchers considered the material that had the better output voltage for various pressures applied and a better sensitivity, thus requiring less mechanical stress to give the highest possible output. PZT, with the chemical formula $\text{Pb}[\text{Zr}_x\text{Ti}_{1-x}]\text{O}_3$, showed great capabilities based on testing conditions. PZT has a relatively high mechanical coercive field, making it promising for high-voltage applications. PZT has different grades (PZT-4, PZT-5A, PZT-5H and PZT-8), and the choice of a suitable grade was largely dependent on the amount of stress to be experienced. Due to their high Q_m , PZT-4 and PZT-8 would be more suitable for high stress and friction, like on roads and pavements. For household constructions where less stress is experienced, PZT-5A and PZT-5H would be more suitable.

The preferred piezoceramics (PZT-5A) were fitted beneath a 10 cm by 10 cm tile and 3D printed using the lightest but strongest filament, polycarbonate (PC) [27], [28]. The lighter the tile, the less expensive it is to mass produce and the better the tile—piezoceramics—touch sensitivity. All of this would improve the efficiency of the system.

The web application received the data transmitted by the ESP32, tracked foot traffic, and recorded energy production on a daily basis. In place of the web application, for general tracking of voltage and current values, the Arduino IDE serial monitor could be used. The piezoelectric cells-circuit was connected in series to get more voltage. The circuit connected the two sensors to the ESP32 on the breadboard. An ESP32 is a low-power system on a chip microcontroller with integrated Wi-Fi and dual-mode Bluetooth. This device is the one that will transmit data to be displayed on a web application. The ESP32 was programmed using Arduino IDE to calculate the power from the current and voltage produced and then transmit the results to the web application. The web application was designed to keep track of data on foot traffic, energy produced daily, and arithmetic operations like the calculation of the amount of energy that can be expected per year. Furthermore, the web application could be accessed remotely using the IoT. In the near future, this web application will be used to monitor surveillance cameras too to combat theft.

Experimental setup

The experimental design was focused on simulating a real-world scenario involving human footsteps to test the hypothesis coupled with the theoretical calculations of predicted outcomes. The designed 10 cm by 10 cm tile was 3D printed using a polycarbonate filament, with PZT piezoceramics in a series-parallel circuit (Fig. 3). The tile converted dynamic energy from any form of applied force, which was to be stored in a 9 V lithium polymer battery (Fig. 4). A lithium polymer battery was chosen because it is an efficient charger, easy to dispose of and has a high energy density [29].

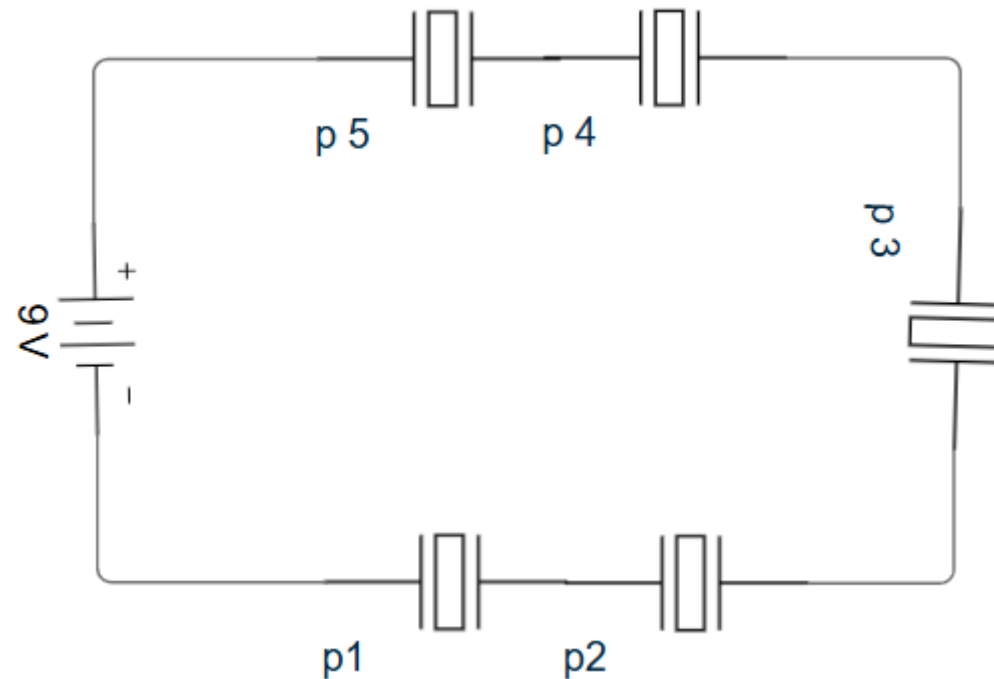


Figure 3 – Schematic diagram of initial piezoceramic series circuit. Showing piezoceramics p1, p2, p3, p4, and p5 in series, connected to a 9V battery.

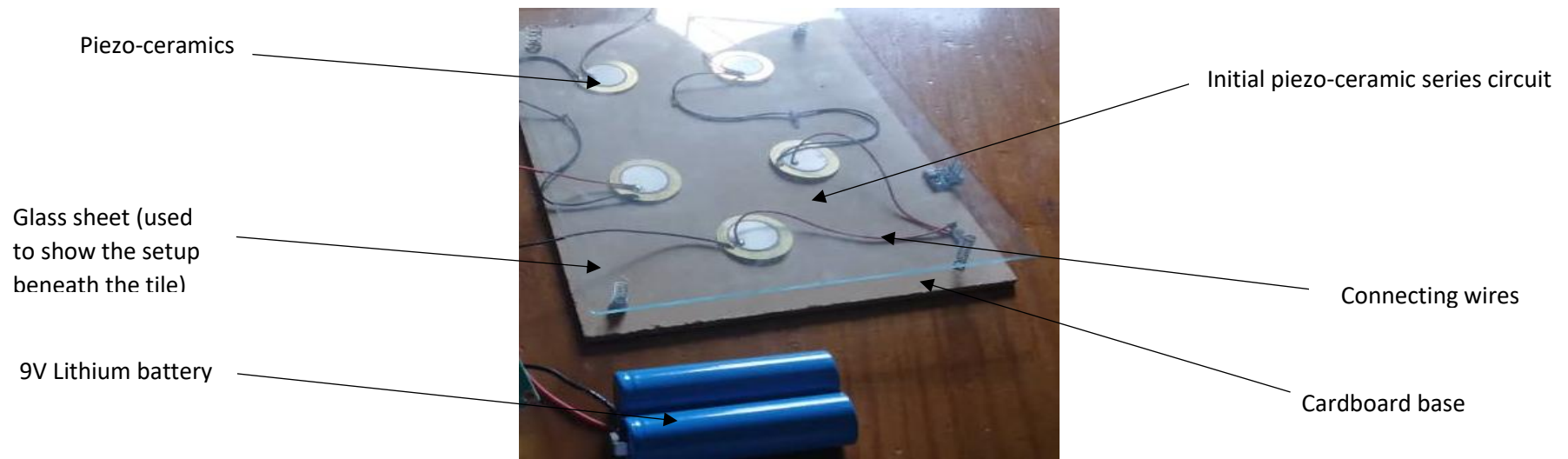


Figure 4 – Initial piezoceramics series circuit. Comprised of PZT piezoceramics, 9 V lithium battery, glass sheet and a cardboard base.

Stored energy was used to power an ESP32 (a microcontroller with Wi-Fi capabilities), a current sensor, a voltage sensor, and a load in the form of an LED connected to the current sensor in order to measure the current produced (Fig. 5). The components used to build this prototype were purchased at Netro Electronics, a robotics and electronics store based in Harare, Zimbabwe.

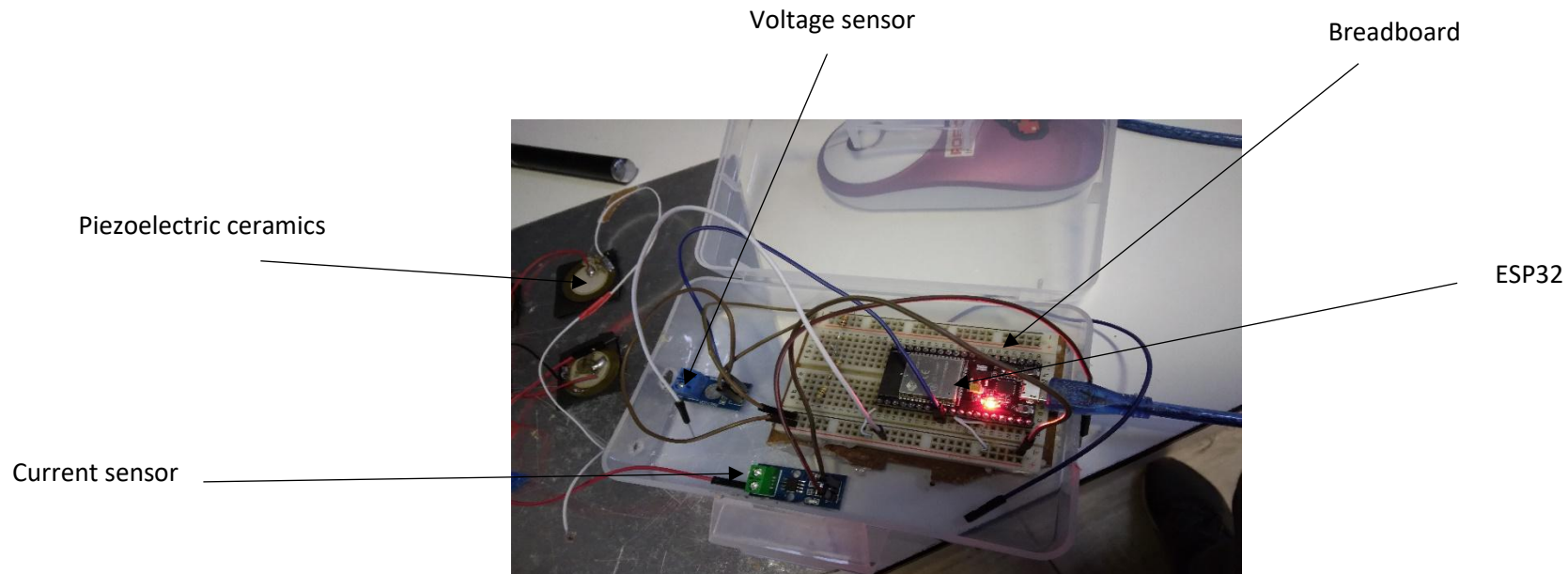


Figure 5 – A prototype circuit on a breadboard. The current sensor, voltage sensor and ESP32 were used to produce a current.

Assembly process

Piezoelectric Series Connection:

Five piezoelectric cells were connected in series to enhance the voltage output. In a series connection, the positive terminal of one cell was connected to the negative terminal of the next, ensuring that the total voltage output increased with each additional cell; the output terminals of the piezoelectric series circuit were wired to both the current and voltage sensors (Fig. 6).

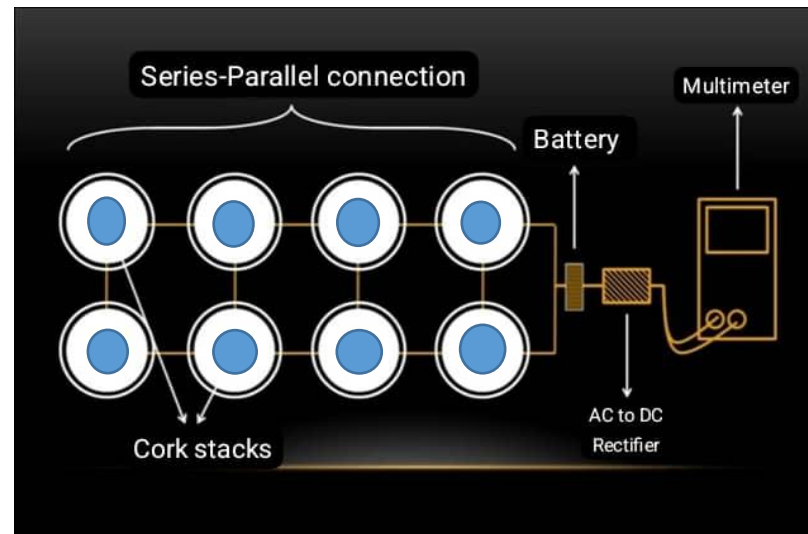


Figure 6 – A circuit diagram. The piezoelectric ceramics were arranged in a series-parallel connection to ensure maximum voltage output. The output terminals of the circuit were wired to the current and voltage sensors.

To increase the voltage output, the preferred piezoelectric ceramics were connected in series. The series-parallel configuration helped manage the behavior of piezoelectric elements and avoided the scenario where they cancel each other out, which would occur if they were connected inappropriately (with opposing polarities causing interference) [30]. When piezoelectric elements were not all pressed simultaneously, connecting them in a parallel or series-parallel configuration could prevent cancellation of the electrical charges produced by different

elements. This way, the voltages across the elements remained constant while the currents summed together. This ensured that if one element generated a positive charge, it would not cancel out another element's negative charge. In a series configuration, the voltage across each element added up, while the current remained the same throughout the circuit. As varying forces were applied to the connection, corresponding voltages were noted.

Connection of Sensors to ESP32:

Current Sensor – The current sensor measured the electrical current generated by the piezoelectric cells. It was connected to an analogue pin on the ESP32 via the breadboard.

Voltage Sensor – Similarly, the voltage sensor measured the voltage produced by the piezoelectric series circuit and it was connected to another analogue pin on the ESP32.

These sensors continuously monitored the output, enabling real-time measurement of energy generation.

ESP32 Setup and Data Transmission:

The ESP32 was placed on the breadboard and connected to both the current and voltage sensors using jumper wires (with female to female pins).

It was programmed using the Arduino IDE to calculate the power generated by the piezoelectric cells using the formula:

$$\textbf{Power (P) = Current (I) \times Voltage (V)}$$

The ESP32 was configured to transmit this data wirelessly (via Wi-Fi) to a web application.

TESTING AND RESULTS

It is important to note that the dependent variables used throughout the testing process of this paper included voltage, current and energy stored. Likewise, the independent variables were the number of steps recorded.

The researchers did not get the exact anticipated voltage value (12 V) during the testing process of the prototype by using several forces and footsteps; however, by making necessary adjustments such as running more trials and using varying types of materials, the small difference of 0.6 V could be eliminated.

Force Application

Researcher 1 and Researcher 2, with masses of 64 kg and 58kg respectively, applied force to the designed tile by taking several steps. The average force exerted was calculated using the equation:

$$\mathbf{F} = \mathbf{m} \times \mathbf{g}$$

Where $\mathbf{F}$ is the force in Newtons (N), $\mathbf{m}$ is mass in kilograms (kg), and $\mathbf{g}$ is the gravitational acceleration, with a value of 9.8 m/s².

For Researcher 1:

$$\mathbf{F} = 64 \times 9.8 = 627.2 \text{ N}$$

For Researcher 2:

$$\mathbf{F} = 58 \times 9.8 = 568.4 \text{ N}$$

The direct piezoelectric effect was used to calculate the amount of charge generated from the applied mechanical force. The direct effect was preferred over the inverse effect because the main focus was on finding out how much charge can be generated by applying mechanical force, whereas using the inverse effect would unnecessarily involve applying an electrical field to produce mechanical deformation. The direct effect relationship is given by the equation:

$$P = d \times \sigma$$

The value of the piezoelectric coefficient (d) that was used for PZT-5A is 390×10^{-12} Coulombs per Newton (C/N).

The mechanical stress (σ) was calculated by dividing the biggest force by the area over which it was applied. The tile's surface area was 0.01m^2 ($0.1\text{m} \times 0.1\text{m}$). Using Researcher 1's force of 627.2 N:

$$\sigma = \frac{627.2 \text{ N}}{0.01 \text{ m}^2} = 62,720 \text{ N/m}^2$$

Thus, the polarization generated was:

$$P = (390 \times 10^{-12} \text{ C/N}) \times 62,720 \text{ N/m}^2 = 24.4 \times 10^{-9} \text{ C/m}^2$$

The coupling coefficient K_{33} for the piezoelectric material was calculated as:

$$K_{33} = \frac{d_{33}}{\sqrt{S_{33} e_{33}}}$$

where

$$d_{33} = 390 \times 10^{-12} \text{ C/N [31]}$$

$$S_{33} = 16.5 \times 10^{-12} \text{ C/N [31]}$$

$$e_{33} = 1.503 \times 10^{-8} \text{ C/N [31]}$$

$$\therefore K_{33} = \frac{390 \times 10^{-12}}{\sqrt{(16.5 \times 10^{-12})(1.503 \times 10^{-8})}} \approx 0.63$$

Results

Multiple trials were conducted, where each researcher applied approximately 10 steps per trial. 10 was the standard number of steps chosen in this experiment to ensure calculations were not too bulky. The force was also varied by introducing objects of different masses to test the tile's response to varying mechanical stress levels.

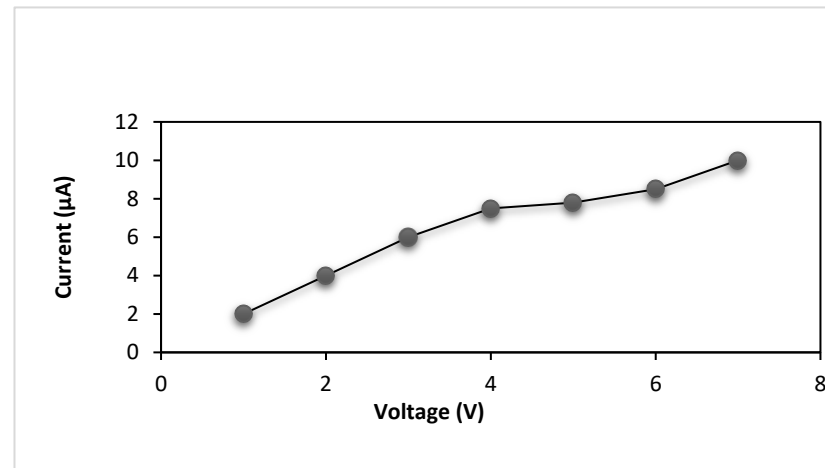


Figure 7 – Results obtained of Current vs. Voltage.

Three trials were taken at each voltage point to find the value of the mean of each electric current (Table 3). A multi-meter was used to take these readings.

Table 3 – Mean values of current readings taken over 3 trials.

Trial 1 (μA)	Trial 2 (μA)	Trial 3 (μA)	Mean Current (μA)
1.8	2.2	2.0	2.0
3.9	4.2	3.9	4.0
5.7	6.3	6.0	6.0
7.2	7.7	7.6	7.5
7.5	7.9	8.0	7.8
8.3	8.6	8.5	8.4
9.8	10.2	10.0	10.0

The mean values were used to calculate the standard deviation (***SD***), and then subsequently calculate the standard error of the mean (***SEM***) (Table 4). Calculating both the ***SD*** and the ***SEM*** was critical for assessing the consistency of data, comparing datasets, and evaluating the reliability of the mean.

$$SD = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Where x_i = each data point, $\bar{x}$ = mean of the data, and n = number of data points. From that:

$$SEM = \frac{SD}{\sqrt{n}}$$

It is important to note that all statistical calculations made in this section were calculated using Microsoft Excel 365 (2021 version).

Table 4 – Mean current values, standard deviations, and standard errors of the mean (SEM) at various voltage levels from 3 trials.

Mean Current (μA)	Standard Deviation (μA)	Standard Error of the Mean (SEM)
2.00	0.20	0.115
4.00	0.17	0.100
6.00	0.30	0.173
7.50	0.26	0.153
7.80	0.26	0.153
8.47	0.15	0.088
10.00	0.20	0.115

The SEM values ranged from 0.088 μA to 0.173 μA , indicating relatively low variability in the mean current readings across the three trials at each voltage level. This suggests that the experiment produced consistent results, and the measurements of current at each voltage level were reliable. It could also be noted that the SD values were low (ranging between 0.15 μA and 0.30 μA), indicating that the individual current readings for each voltage level did not deviate significantly from the mean. However, there is a slightly larger variability at higher voltage levels like 6 V, as indicated by the higher SEM and SD values. This was mainly due to the physical properties of the piezoelectric materials and could be reduced by running more trials.

There was a variation of the current (μA) and voltage (V) due to the series circuit connection of the Piezoceramics (Fig. 6). It was noted that as voltage increased, current did too, although there were some notable variations in the increase. The testing results also depended on the ceramics' polycrystalline structure which consists of numerous crystallites and a polarity of elementary cells. The anticipated results differed slightly from the outcome because even though the piezoceramics were all composed of PZT material, they still had different properties due to their polycrystalline makeup [29]. A single piezoelectric ceramic produced varying voltages between 0.7 V and 2.5 V, mainly due to the amount of force exerted and the crystalline properties of the ceramics. The results from this graph were crucial in calculating the amount of electric power that could be obtained from the current and voltage values, which subsequently helped determine the amount of electric energy generated over a certain period. Electric power could be calculated using the formula:

$$\text{Power (P)} = \text{Current (I)} \times \text{Voltage (V)}$$

The electric energy produced could then be calculated using the formula:

$$\text{Energy (E)} = \text{Power (P)} \times \text{Time (t)}$$

Employing a series-parallel combination increased both the total current and voltage, which in turn gave more energy.

Different objects with different masses (which in some cases were heavier than an average adult male in Zimbabwe) were tested on the tile to get the highest possible output. However, to ensure critical practicality and credibility at the initial stage, only the results obtained from the forces exerted as the Researchers stepped on the tile were considered, since this research was primarily focused on human footsteps (Fig. 8).

From the force of 627.2 N ($64 \text{ kg} \times 9.8 \text{ m/s}^2$) applied by Researcher 1, a single smart tile with 5 piezo ceramics beneath it produced 11.4 V. Although it did not reach the projected outcome of exactly 12 V, adjustments could be made to get the 12V or even more. This proved the practicality of the hypothesis tested to generate electricity for basic needs like lights and charging phones in communities like Dzivarasekwa (a suburb of Harare, Zimbabwe).

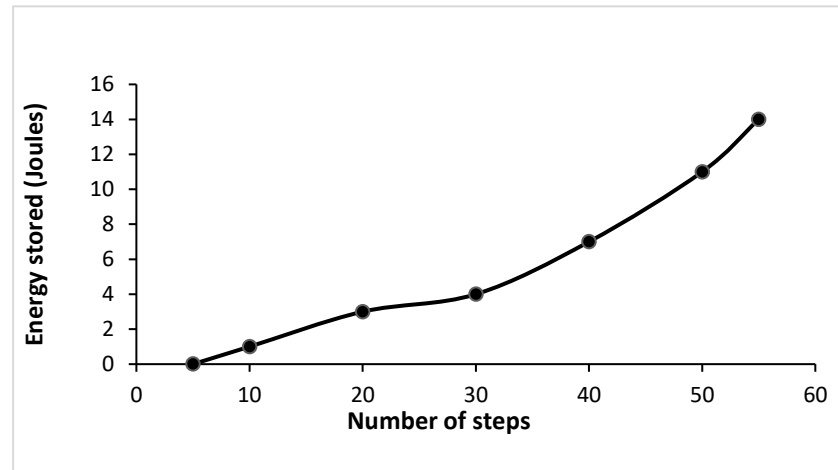


Figure 8 – Diagram of Energy stored vs. Number of steps.

SUMMARY OF FINDINGS

The Electric Step was compared to other conventional methods of electricity generation, as well as other forms of piezoelectric technology [32], [33].

Table 5 – Comparison of The Electric Step to other electricity generation methods. The comparison focused on cost-effectiveness, availability in Africa, environmental impact, reliability and scalability.

Aspect	The Electric Step	Fossil Fuels (Coal, Oil, Gas)	Hydroelectricity	Renewable energy (Solar, wind)	Nuclear	Other Piezoelectric Technologies
Cost-effectiveness	Low cost (in the long-term)	High cost	High Cost	Varies	Varies	Varies
Availability in Africa	Abundant resources	Limited	Limited	Varies	Varies	Potential
Environmental impact	Clean energy	Pollution	Environmental impact	Clean energy	Clean energy	Clean energy
Reliability	Depends on foot traffic	Reliable	Reliable	Depends on source	Reliable	Depends on application
Scalability	Scalable	Limited	Limited	Varies	Varies	Varies

Cost-effectiveness

The Electric Step is a cost-effective solution in the long-term, utilizing existing foot traffic and 3D technology to generate energy, thereby mitigating the need for costly long-term infrastructure investments. Traditional methods like fossil fuels (coal, oil and gas) entail high costs associated with extraction, transportation, and refining processes. Hydroelectricity, another conventional approach, demands substantial initial investments for dam construction and related infrastructure. Meanwhile, the cost of renewable energy sources (solar and wind) fluctuates depending on factors like geographical location, technological advancements, and governmental incentives. Nuclear energy, although clean, entails notable upfront expenditures for the establishment and maintenance of nuclear power plants. Additionally, the cost of other piezoelectric technologies varies contingent upon the specific application and technology employed.

Availability in Africa

The Electric Step represents a promising avenue for Africa, given its abundant resources for implementing piezoelectric technology, particularly in areas with high foot traffic. On the other hand, the continent's reliance on fossil fuels is constrained by limited availability compared to other regions. Similarly, Africa's potential for hydroelectricity is restricted relative to other continents, highlighting the need for diversified energy sources. Renewable energy offers varying potential across Africa, contingent upon geographic location and climatic conditions. Meanwhile, the accessibility of nuclear energy hinges on factors like existing infrastructure, investment capabilities, and geopolitical considerations. The availability of other piezoelectric technologies is subject to specific application and resource availability, warranting careful consideration in energy planning and implementation strategies.

Environmental Impact

The Electric Step, powered by piezoelectric technology, stands out for its ability to generate clean energy without polluting the environment or contributing to environmental degradation. In contrast, fossil fuels are notorious for their pollution and greenhouse gas emissions, exacerbating environmental degradation and climate change. While hydroelectricity is considered renewable, the construction of dams poses great environmental risks, impacting ecosystems and local communities. Similarly, solar and wind energy offers clean and renewable alternatives with minimal environmental impact compared to fossil fuels. Nuclear energy boasts clean emissions but raises concerns regarding nuclear waste disposal and the potential for accidents. Additionally, the environmental impact of other piezoelectric technologies varies depending on the specific application and materials utilized, necessitating careful consideration in energy planning and implementation.

Reliability

The reliability of the Electric Step, utilizing piezoelectric technology, hinges on the intensity of foot traffic and the durability of the materials employed in its construction. In contrast, conventional methods like fossil fuels and nuclear energy typically exhibit reliability once the necessary infrastructure is established. Hydroelectric power is renowned for its reliability, yet its availability may be susceptible to factors such as droughts or seasonal variations in water flow. Conversely, the reliability of renewable energy sources is contingent upon weather

conditions, which can fluctuate over time and across locations. The reliability of other piezoelectric technologies is subject to the specific application and technological advancements utilized, warranting careful assessment and consideration in energy planning and implementation strategies.

Scalability

Piezoelectric technology, exemplified by the Electric Step, offers scalability and adaptability, allowing for its implementation across diverse settings to harness energy from various sources of mechanical stress. In contrast, the scalability of conventional methods like fossil fuels and nuclear energy may be limited by factors such as resource availability and environmental considerations. While hydroelectric power is scalable, its expansion is constrained by the availability of suitable river locations and concerns about environmental impacts. Additionally, the scalability of solar and wind energy hinges on factors like land availability, advancements in technology, and the development of grid infrastructure. The scalability of other piezoelectric technologies is dependent upon the specific application and technological advancements utilized, highlighting the importance of careful assessment and planning in energy implementation strategies.

In summary, the comprehensive comparison of different energy sources (Table 5) shows their unique characteristics and implications for energy generation in Africa.

DISCUSSION

Graphs and mechanical force

Although the anticipated voltage output of 12 V was not achieved, the experimental setup yielded a maximum voltage of 11.4 V when a force of 627.2 N was applied. This demonstrated the potential of the system to generate significant power with necessary adjustments. This demonstrated the potential of the system to generate significant power with necessary adjustments. Due to outside factors like loss of energy through heat, resistance, and building material used, current levels from the Current vs. Voltage graph (Fig. 7) varied notably well, but this was due to the voltage being lower than the current (as indicated by the graphical trend). This problem could be improved by implementing a series-parallel connection where both a higher voltage and a higher current can be obtained due to the circuit's ability to increase or decrease the overall resistance. The sum of individual voltages in series is equivalent to the total voltage source of the circuit. As the number of connected branches increases in a parallel circuit, the current increases: thus, combining series and parallel creates a balance of both current and voltage. This would allow for optimization of the circuit for specific requirements, such as increasing the current capacity or reducing energy consumption

One of the main reasons for the variation in voltage output was attributed to the polycrystalline structure of piezoelectric ceramics used. Despite being made from the same PZT-5 material, the ceramics exhibited different properties. Factors such as the level of polarization influence the difference in voltage: a less-polarized material would generate a weaker piezoelectric response, which would directly impact voltage output. This explained the range of voltages (0.7 V – 2.5 V) generated by individual ceramics depending on the force applied. The variations of forces applied also played a part in the overall voltage obtained. The introduced different masses of 64 kg and 58 kg resulted in forces of 627.2 N and 568.4 N respectively, which in turn produced different amounts of charge and voltage. The output of the piezoelectric material was not a steady one, thus a bridge circuit should be used to convert this variable voltage into a linear one. The voltage produced across the tile can be seen on a liquid crystal display (LCD) or serial monitor.

Furthermore, the coupling coefficient K_{33} for the piezoelectric material, calculated as 0.63, showed the efficiency of energy transfer in the system. This value, which was calculated from the piezoelectric coefficient, mechanical compliance and permittivity, confirmed the suitability of PZT-5A ceramics for energy harvesting applications. These results determined that using additional piezoelectric ceramics, increasing their size, and conducting further trials with different materials could lead to higher overall power output and improve system performance.

Table 3 shows 3 trials taken using a multi-meter at each voltage point to find the value of the mean of each electric current. The purpose of finding the mean values was to use it to calculate the Standard Deviation, and then subsequently calculate the Standard Error of the Mean (SEM). Calculating both the Standard Deviation and the Standard Error of the Mean is critical to this paper, because it aids in assessing the consistency of data, compare datasets, and evaluate the reliability of the mean. The SEM values in Table 4 range from 0.088 μA to 0.173 μA , indicating relatively low variability in the mean current readings across the three trials at each voltage level. This suggests that the experiment produces consistent results, and the measurements of current at each voltage level are reliable. It can also be noted that the SD values are low (ranging between 0.15 μA and 0.30 μA), indicating that the individual current readings for each voltage level do not deviate significantly from the mean. There is a slightly larger variability at higher voltage levels like 6V, as indicated by the higher SEM and SD values, however the variability appears to be smaller at higher voltage levels like 8.47 and 10, this may be due to the system having nonlinear characteristics at some parts of the tile.

Indicated in Figure 8 is the variation of Current in micro Amps and Voltage in Volts due to the series circuit connection of the piezo-ceramics. It can be noted that as Voltage increased, current did too, although there were some notable variations in the increase. The testing results also depended on the ceramics' polycrystalline structure which consists of numerous crystallites and a polarity of elementary cells. The anticipated results differed slightly from the outcome because even though the piezo ceramics were all composed of PZT material, they still had different deformation properties due to the polycrystalline makeup [28]. From readings taken by a multi-meter, a single piezoelectric ceramic produced varying voltages between 0.7V-2.5V mainly due to the amount of force exerted and the crystalline properties of the ceramics. The results from this graph are crucial to calculate the amount of electric power that can be obtained from the Current and Voltage values which subsequently helps determine the amount electric energy generated over a certain period. Electric power can be calculated using the formula: Power (P) = Current (I) $\times$ Voltage (V), and electric energy produced can be calculated using the formula: Energy (E) = Power (P) $\times$ Time (t). Employing a Series-Parallel combination will increase both the total Current and Voltage, which in turn give more energy.

Limitations

Energy is converted into other non-useful forms (electrical and mechanical losses during conversion) and the output of the piezoelectric material is not always a steady one, making it difficult to predict the amount of energy that can be produced over a long period or annually. Furthermore, the web application still needs to be constantly updated to suit the needs of day-to-day applications.

The practical and commercial implementation of the smart energy system should have the ability to suit real-life situations, so all possible outcomes must be explored if the smart tiles are to be deployed on sidewalks and other major areas. Smart tiles are prone to vandalism or theft: to better prepare for these scenarios, it would be ideal to have surveillance cameras set up in the surrounding areas. The use of recycled material poses a great relief on the energy usage during the making of floor tiles similar to the ones constructed in homes. The tiles can be retrofitted to existing constructions or can be planned into any new ground surface framework easily. Elimination of such limitations means the tiles can be designed in any shape to meet the required needs.

LONG-TERM USAGE PLAN

Implementation and Deployment

Smart floor tiles will be implemented in phases, starting with pilot installations in selected urban and rural areas. These installations will serve as test cases to evaluate the performance and feasibility of the technology in real-world conditions. Over the next few years, the project will be scaled up gradually to cover more communities across Zimbabwe and other parts of Africa. Based on Zimbabwe's historical energy background and assumptions of projected growth like the installation of smart floor tiles in areas with high foot traffic (urban centres, schools, or transportation hubs), it is predicted that the projected energy generation will increase exponentially with time (Fig. 8). Foot traffic data is crucial to understand the future of The Electric Step. Urban areas with dense populations, such as city centres, train stations, and markets, have been tracked by urban planning and transportation departments in Harare, Zimbabwe [32].

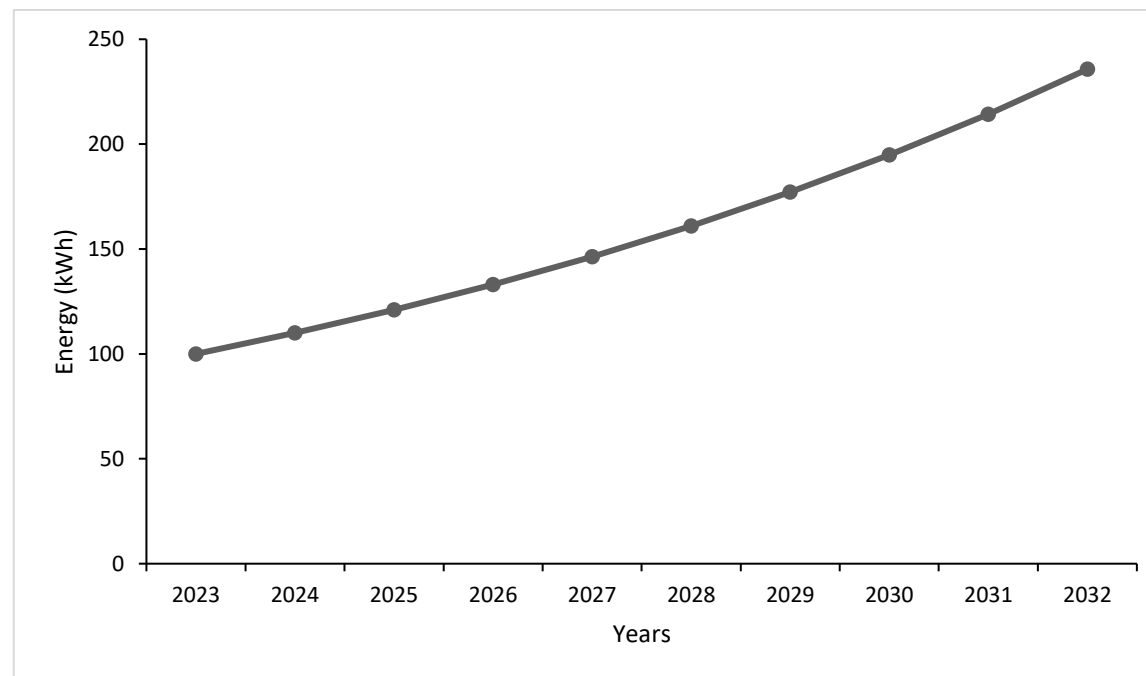


Figure 9 – Projected energy generation growth in Zimbabwe. Based on the country's historical energy background

Partnerships and Collaboration

To ensure the success and sustainability of the project, partnerships will be established with local governments, non-profit organizations, and private sector stakeholders. These partnerships will help provide funding, technical expertise and support for infrastructure development and maintenance.

Cost savings (USD)

In discussing the long-term usage plan of The Electric Step, the importance of financial efficiency in energy initiatives cannot be ignored. Aligning with the project goals, an approach was outlined through estimated calculations on the potential cost savings from implementing smart floor tiles compared to traditional energy sources over one year (Table 6). At the initial implementation stage, a pilot test will be carried out over this period, involving the construction of 100 tiles in a single house in Dzivarasekwa. The performance of the tiles will be monitored over 12 months to analyse the cost savings associated with the new system compared to the traditional systems currently in use.

Table 6 – Cost savings data on The Electric Step.

Energy Source	The Electric Step
Initial Investment (\$)	100,000 (for equipment, installation and monitoring systems)
Cost per kWh (\$)	0.10 (assuming the electricity tariff)
Annual maintenance cost (\$)	5,000
Estimated generation per year (kWh)	50,000

Initial investment amortized over 10 years (\$)	10,000 (100,000 / 10)
Annual operational cost (\$)	5,000
Total annual cost (\$)	15,000 (10,000 + 5,000)

The cost per kWh for traditional energy sources such as coal-fired power plants varies widely depending on factors such as location, infrastructure, fuel prices and government regulations. To estimate the cost per kWh of these energy sources, a hypothetical value of \$0.15/kWh was used (Table 7). This value was within the typical range for commercial electricity tariffs in many regions. This value was based on general knowledge and common electricity prices observed in various region [35].

Table 7 – Cost savings data for traditional energy sources.

Energy source	Traditional energy sources (e.g. hydroelectricity, coal)
Cost per kWh (\$)	0.15 (Higher than The Electric Step due to operational costs)
Annual electricity consumption (kWh)	50,000
Total annual cost (\$)	7,500 (50,000 kWh × \$0.15)

Electricity tariffs

The tariff for traditional energy sources is approximately \$0.15/kWh, which falls within the global average range for coal and hydroelectric plants. The higher operational costs that these traditional sources often incur are due to processes like fuel extraction, transportation, and regulatory fees (such as emissions compliance), which raise the overall price per kWh. A \$0.05 difference between The Electric Step and traditional sources appears marginal at first glance, but can accumulate significantly over time. For example, with annual electricity consumption set at 50,000 kWh, this difference results in a \$2,500 price gap (\$5,000 for The Electric Step versus

\$7,500 for traditional sources). The annual cost savings can be recorded by subtracting the annual cost for The Electric Step (Table 3) from the annual cost for traditional energy sources (Table 4) to give -\$7,500 (\$7,500-\$15,000).

The negative cost savings indicate that The Electric Step has higher initial and operational costs compared to traditional energy sources. However, it is essential to consider the broader context and long-term implications. In the short term, compared to the Electric Step, traditional energy sources may tend to benefit more or be less expensive due to their already existing infrastructure. However, these savings are often offset by fuel and environmental costs.

In the long term, as maintenance and fuel costs continue to rise for traditional energy sources, The Electric Step can potentially offer greater savings and reliability. In regions where energy infrastructure is underdeveloped or where foot traffic is consistently high, The Electric Step can become particularly cost-effective. The estimated operational cost of \$5,000 per year for The Electric Step remains predictable, while traditional systems face higher volatility depending on global fuel markets.

While The Electric Step may incur higher upfront costs or ongoing expenses, it offers numerous benefits beyond immediate financial savings. These benefits include environmental sustainability, reduced reliance on finite resources, and long-term resilience against fluctuating energy prices. Therefore, despite the higher cost, The Electric Step's overall value proposition should be evaluated based on its holistic impact and contribution to the UN SDGs.

Community Engagement and Education

Community engagement and education programs will be implemented to raise awareness about the benefits of the technology and promote its adoption. Workshops, seminars, and outreach campaigns will be organized to educate residents about energy conservation and sustainability, and the role they can play in contributing to the project's success.

Monitoring and Evaluation

A comprehensive monitoring and evaluation system will be established to track the performance and impact of the project over time. Key performance indicators such as energy generation, cost savings and community satisfaction will be regularly monitored and analysed to assess the project's effectiveness and identify areas for improvement.

Expansion and Innovation

As the project matures, efforts will be made to explore new applications and innovations that can further enhance its impact and reach. This may include integrating additional features or functionalities into the existing infrastructure, exploring new materials or technologies for energy generation, and expanding the project's scope to cover other related areas such as water conservation or waste management.

Sustainability and Maintenance

To ensure the long-term sustainability of the project, mechanisms will be put in place for ongoing maintenance and upkeep of the infrastructure. This may involve training local technicians or community members to handle routine maintenance tasks, establishing service contracts with third-party providers, and implementing revenue-generating models to cover operational costs.

CONCLUSION

Overall, the testing confirmed the hypothesis that piezoelectric ceramics can effectively convert mechanical energy from human footsteps into significant electrical energy. Although some discrepancies were observed between the theoretical and experimental results, namely the slight difference in the theoretical voltage of 12 V and the practically tested voltage of 11.4 V, the system's ability to generate near-expected voltages highlights its potential for practical use because 11.4 V can still power lights and other small devices in a community like Dzivarasekwa in Harare, Zimbabwe. The Electric Step provides potential long-term benefits such as environmental sustainability, energy independence and resilience against fluctuating energy prices. While initial costs are higher than traditional energy sources, partnerships with local governments, non-profit organizations and private stakeholders will help secure funding, technical expertise, and support for infrastructure development. Over time, as fuel prices rise and environmental considerations become more pressing, the economic advantages of The Electric Step are expected to increase. This project was designed to potentially decrease energy poverty in Africa: it is a solution for the people, and it is critical to educate the people on how to effectively work towards sustainable energy through the Electric Step.

References

- [1] Harvey, "African Development Report," Oxford University Press, New York, 2007.
- [2] C. Carmona, "Energy resources," 11 March 2017. [Online]. Available: <https://aquaknow.jrc.ec.europa.eu/en/wefe-nexus/documents/atlas-africa-energy-resources>. [Accessed 11 May 2022].
- [3] I.D. Ibrahim, Y. Hamam, Y. Alayli, T. Jamiru, E.R. Sadiku, W.K. Kupolati, J.M. Ndambuki, "A review on Africa energy supply through renewable energy production," *Energy Strategy Reviews*, vol. 38, p. 3, 2021.
- [4] World Bank, "World Bank," August 2023. [Online]. Available: <https://www.worldbank.org/en/region/afr/overview>. [Accessed 31 August 2024].
- [5] I. Nhire, "Green Business Gazette," August 2022. [Online]. Available: <https://www.gbg.co.zw/2022/12/13/can-africa-win-from-exploiting-its-oil-and-gas/>. [Accessed 2 September 2024].
- [6] A. O. a. M. Maiga, "The socio-economic impacts of renewable energy deployment in sub-Saharan Africa: A case study of solar energy," *Journal of Renewable Energy Research*, vol. 13, no. 4, pp. 1578-1590, 2023.
- [7] U. N. D. Programme, "Human Development Report 2023: Beyond income, beyond averages, beyond today: Inequalities in human development in the 21st century," UNDP, 2023.
- [8] "Sustainable Energy for all," 31 March 2021. [Online]. Available: <https://www.seforall.org/news/health-facility-electrification-in-sub-saharan-africa>. [Accessed 5 June 2023].
- [9] P. O. D. T. Andries Rossouw, "Africa Energy Review 2021," PwC, Lagos, 2021.
- [10] "Harnessing Geothermal Energy in Africa," African Union, Addis Ababa, 2020.
- [11] E. M. A. T. T. Alli Dimple Mukasa, "Development of wind energy in Africa," African Development Bank Group, 2012.
- [12] M. Taylor, "Solar PV in Africa: Falling costs driving rapid growth," *ESI Africa*, no. 5, 2016.

- [13] M. W. Nganga, "World Economic Forum," 17 November 2016. [Online]. Available: <https://www.weforum.org/agenda/2016/11/understanding-africas-energy-needs/>. [Accessed 15 May 2022].
- [14] R. Shirley, "The Conversation," 19 April 2018. [Online]. Available: <https://theconversation.com/millions-of-urban-africans-still-dont-have-electricity-heres-what-can-be-done-92211>. [Accessed 1 June 2023].
- [15] M. P. B. a. M. Cosgrove-Davies, "Electricity Access in Sub-Saharan Africa," Agence française de développement and the World Bank , 2019.
- [16] Sona Solar Zimbabwe, "Polaris 12V 100Ah Lithium Ion Battery," [Online]. Available: https://www.sonasolar.co.zw/2024/03/polaris-12v-100ah-lithium-battery.html?utm_source=chatgpt.com. [Accessed 13 December 2024].
- [17] J. M. L. a. D. J. I. Henry A. Sodano, "An experimental comparison between several active composite actuators for power generation," *SPIE. Digital library*, vol. 5390, 2004.
- [18] T. Ikeda, in *Fundamentals of Piezoelectricity*, Oxford, Oxford University Press, 1996, p. 45.
- [19] K. Uchino, in *Piezoelectric Actuators and Ultrasonic Motors*, Boston, Kluwer Academic Publishers, 1997, p. 123.
- [20] T. Ikeda, *Fundamentals of Piezoelectricity*, Oxford: Oxford University Press, 1996.
- [21] S. B. a. J.-J. C. C Jean-Mistral, "Comparison of electroactive polymers for energy scavenging applications," *Smart Materials and Structures*, vol. 19, no. 8, 2010.
- [22] K. Uchino, in *Piezoelectric Devices and Materials*, Hoboken, John Wiley & Sons, 2010, p. 150.
- [23] K. Uchino, in *Piezoelectric Devices and Materials*, Hoboken, John Wiley & Sons, 2010, p. 153.
- [24] H. R. A. H. Y. J. S. Gao, "7th Annual International Conference on Materials Science and Engineering," *IOP Conference Series: Materials Science and Engineering*, vol. 562, p. 012098, 2019.
- [25] H. Liu, J. Zhong, C. Lee, S.-W. Lee and L. Lin, "A comprehensive review on piezoelectric energy harvesting technology: Materials, mechanisms, and applications", *Applied Physics Reviews*, vol. 5, no. 4, 2018.

- [26] P. S. L. Tao Li, "Piezoelectric Energy Harvesting Technology: From Materials, Structures, to Applications," *Small Structures*, vol. 2, no. 3, p. 3, 2021.
- [27] S. Cahoon, "MatterHackers," 22 August 2017. [Online]. Available: <https://www.matterhackers.com/articles/how-to-succeed-when-printing-with-polycarbonate-filament>. [Accessed 17 February 2024].
- [28] T. Bardwell, "3DSourced," 25 June 2023. [Online]. Available: <https://www.3dsourced.com/3d-printer-materials/polycarbonate-pc-filament-3d-printing/>. [Accessed 28 February 2024].
- [29] J. M. E. S. S. C. K. W. P. J. T. P. Gyo Hun, "Introducing the energy efficiency map of lithium-ion batteries," *International Journal of Energy Research*, vol. 43, no. 2, pp. 931-944, 2019.
- [30] J. M. G. Farouk Balouchi, "Design and optimisation of a footfall energy harvesting system," *Journal of Intelligent Material Systems and Structures*, vol. 25, no. 14, pp. 1746-1756, 2014.
- [31] PI Ceramic GmbH, "Piezo Material Properties Data Sheet," 12 November 2020. [Online]. Available: <https://info.piezo.com/hubfs/Data-Sheets/piezo-material-properties-data-sheet-20201112.pdf>. [Accessed 28 December 2024].
- [32] A. C. V. S. D. I. Giambattista Guidi, "Environmental Impact of Electricity Generation Technologies: A Comparison between Conventional, Nuclear, and Renewable Technologies," *Energies*, vol. 16, no. 23, 2023.
- [33] IEA, "Renewables and environmental sustainability," [Online]. Available: <https://www.iea.org/reports/renewables-and-sustainability>. [Accessed 18 September 2024].
- [34] Z. N. S. Agency, "Transport statistics report 2018-2022," Harare, 2023.
- [35] "Electricity prices for enterprises worldwide in June 2023, by select country," Statista Research Department, Worldwide, 2024.
- [36] C. Woodford, "explainthatstuff," 21 May 2022. [Online]. Available: <https://www.explainthatstuff.com/piezoelectricity.html>. [Accessed 6 May 2023].
- [37] S. Katzir, *The Beginnings of Piezoelectricity*, Springer Nature, 2006.
- [38] M. Sunar, "Piezoelectric Materials," *Comprehensive Energy Systems*, vol. 2, no. 22, pp. 696-719, 2018.

[39] M. Sunar, "Energy Materials," *Sensors for mechatronics*, vol. 2, no. 22, pp. Pages 696-719, 2018.